(Please write your Enrollment Number)

MID-TERM EXAMINATION

B. Tech. Semester: I

(October, 2025) OFF LINE mode

Subject Code: BAI 101	**Subject: Programming with Python**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mappin**
	a) What is the difference between = and == operators in Python?		**CO1**

	b) What is the output of the following code:
	
	`nums = [5, 10, 15]`
	`result = [x//2 if x > 7 else x+1 for x in nums]`
	`print(result)`
			CO2

	c) What will be the output of the following code:
	
	`text = "bananaapple"`
	`result = set()`
	`for i in range(len(text)):`
	`  if i%2 == 0:`
	`  continue`
	`  result.add(text[i:i+2])`
	`print(result)`
			CO2

	d) Predict the output of the following code:
	
	`count = 1`
	`while count <= 3:`
	`  print("Hi")`
	`  count += 1`
	
	What will happen if we write **while True:** without a break statement?
			CO1

	UNIT I		
Q2	**Attempt any two parts**	**(5*2=10)**	**CO Mappin**
	a) **(i)** Whether the variables and identifiers are same (Y/N)? Justify your answer with suitable example. **a) (ii)** Write a program to print the following pattern: 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5 Assume that the number of rows = 5.	**(2+3)**	**CO1**

	b) (i) Explain identity operator and membership operator with suitable example. **b) (ii)** Write a program to print the Fibonacci series up to a given number n=7.	**(2+3)**	**CO1**
	c) Write a Python program to accept a password from the user and check if it is valid based on the following rules: • At least 8 characters long • Must contain at least one digit • Must contain at least one uppercase letter Print "Valid Password" if all conditions are satisfied, else "Invalid Password". Use if-elif-else with logical operators (and, or) to implement.	**(5)**	**CO1**
	UNIT II	**(5*2=10)**	**CO Mapping**
Q3	**Attempt any two parts**		
	a) Write a program to transpose a given matrix A of size 3x3, subtract the resultant matrix from the original matrix and store the result into third matrix mat-diff, where $A = \begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$	**(5)**	**CO2**
	a) In a company, the details of 5 employees are stored in a dictionary named Emp_details with fields: Emp_id, Name, Age, and Salary. • Two employees have received a salary increment of 10%. Update their salaries in the dictionary and print the updated record. • Print the Emp_id and Name of employees whose salary is now above ₹50,000.	**(5)**	**CO2**
	c) (i) List the mutable and immutable objects. Discuss scenarios with examples where each one is useful. **c) (ii)** Differentiate between ordered and unordered data structures with examples.	**(3+2)**	**CO2**